Given: h(x) = f(g(x)) and h(x) is continuous

x	0	1	2	3
f(x)	2	3	-1	5
g(x)	3	6	0	2

Can you show that h(x)=0 at least once in the interval from 0<x<3.

Because h(x) is continuous and the y values change from positive to negative we know by the IVT that h(x) must cross the x-axis at least once. ie h(x)=0 somewhere in the interval.

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t (minutes)	0	2	5	8	12
$v_A(t)$ (meters/min)	0	100	40	-120	-150

4. Train A runs back and forth on an east-west section of railroad track. Train A's velocity, measured in meters per minute, is given by a differentiable function $v_A(t)$, where time t is measured in minutes. Selected values for $v_A(t)$ are given in the table above.

b) Do the data in the table support the conclusion that train A's velocity is -100 meters per minute at some time t with $5 < t < 8$? Give a reason for your answer.

x	0	1	2	3	4
f(x)	6	-3	5	6	-3

Given that f(x) is continuous, what is the minimum number of times f(x) crossed the x- axis. What theorem did you apply?